A QUASI-COMPACT MORPHISM IS DOMINANT EXACTLY WHEN IT HITS THE GENERIC POINT OF EVERY IRREDUCIBLE COMPONENT OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-quasi-compact-dominant is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 1271-1280 and proof span 1282-1302 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement carries an author slogan ('Morphisms whose image contains the generic points are dominant.'). The project's own shared preamble defines the slogan environment as a comment environment, so the slogan is not typeset, exactly as in the project's own output; it is preserved verbatim in the counted span.

Editorial note. The proof replaces the morphism by one from a finite disjoint union of affine opens; the author's notation for this comparison morphism is preserved.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A disjoint union of finitely many affine schemes is affine, cited from the schemes chapter. Cited at proof lines 1288-1290. Makes the comparison morphism used in the reduction a morphism of affine schemes.
- The algebra criterion that a ring map has dense image exactly when the generic point of every irreducible component of the target lies in the image of the corresponding morphism of spectra. Cited at proof lines 1301-1302. Gives the conclusion in the affine case, to which the proof reduces.

Source reference. Original lemma 00FL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 575–575. Same-chapter reference used in the retained author text at source line(s) 1301, 11842, 11946, 12203, 14779. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01I5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1966–1966. Same-chapter reference used in the retained author text at source line(s) 1294. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact morphism of schemes. Then f is dominant if and only if for every irreducible component $Z \subset S$ the generic point of Z is in the image of f.*

Proof. Let $V \subset S$ be an affine open. Because f is quasi-compact we may choose finitely many affine opens $U_i \subset f^{-1}(V)$, $i = 1, \ldots, n$ covering $f^{-1}(V)$. Consider the morphism of affines

$$f' : \coprod_{i=1,\ldots,n} U_i \longrightarrow V.$$

A disjoint union of affines is affine, see Schemes, Lemma 01I5. Generic points of irreducible components of V are exactly the generic points of the irreducible components of S that meet V. Also, f is dominant if and only if f' is dominant no matter what choices of V, n, U_i we make above. Thus we have reduced the lemma to the case of a morphism of affine schemes. The affine case is Algebra, Lemma 00FL. □